Multidimensional Scaling and Conjoint Analysis

Chapter 26

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Agenda

Overview: two related techniques

Multidimensional scaling (MDS)

Conjoint analysis

Summary and review

Learning objectives

After this session you should be able to:

1. Discuss the basic concept and scope of **multidimensional scaling (MDS)** and its applications in marketing research.
2. Describe the steps of MDS on perception data: formulating the problem, obtaining input data, selecting a procedure, choosing the number of dimensions, labelling the dimensions and assessing reliability and validity.
3. Explain how preference data are scaled, distinguishing **internal** from **external** analysis.
4. Explain **correspondence analysis** and its advantages and disadvantages.
5. Understand how MDS relates to discriminant analysis and factor analysis.
6. Discuss the basic concepts of **conjoint analysis**, contrast it with MDS and describe its applications.
7. Describe how to conduct a conjoint analysis, from formulating the problem to assessing reliability and validity.
8. Define **hybrid conjoint analysis** and explain how it simplifies data collection.

Two techniques for perceptions and preferences

Both techniques start from consumers' *subjective evaluations*, but they answer different questions.

	Multidimensional scaling (MDS)	**Conjoint analysis**
Stimuli	Existing products or brands	Combinations of attribute levels designed by the researcher
Typical input	Similarity judgements, attribute ratings, preference rankings	Ratings or rankings of product profiles
Output	A **spatial map**: brands (and ideal points) located in a low-dimensional perceptual space	**Part-worths** for each attribute level and **relative importance** of each attribute
Key question	How do consumers *see* the market?	Which *trade-offs* do consumers make?

The two are **complementary** and are often used together in the same study.

Example: Colas collide

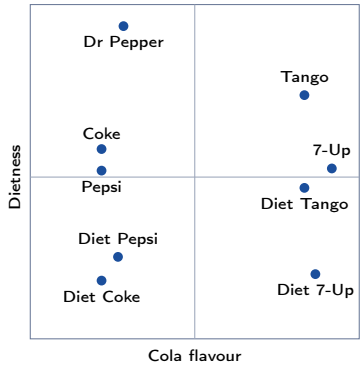
Real research: Colas collide

Participants rank-ordered all possible pairs of nine soft drinks by similarity, and the data were analysed with MDS.

- The horizontal axis was labelled *cola flavour*, the vertical axis *dietness*.
- Coke and Pepsi sit very close together, so they are perceived as near-substitutes. So are 7-Up and Tango, and Diet Coke and Diet Pepsi.
- Dr Pepper sits apart from all the others.

Maps like this show the **competitive structure** of a market at a glance.

Figure 1. Spatial map of soft drinks (illustrative)



Basic concepts in MDS

Multidimensional scaling (MDS)

A class of procedures for representing the **perceptions and preferences** of participants spatially, by means of a visual display. Perceived (psychological) relationships among stimuli are represented as **geometric relationships among points** in a multidimensional space, called a **spatial map**.

The axes of the map are assumed to be the underlying dimensions consumers use to form perceptions and preferences. MDS has been used to identify:

1. the **number and nature of the dimensions** consumers use to perceive brands;
2. the **positioning of brands** on these dimensions;
3. the positioning of consumers' **ideal brand** on these dimensions.

Marketing applications of MDS

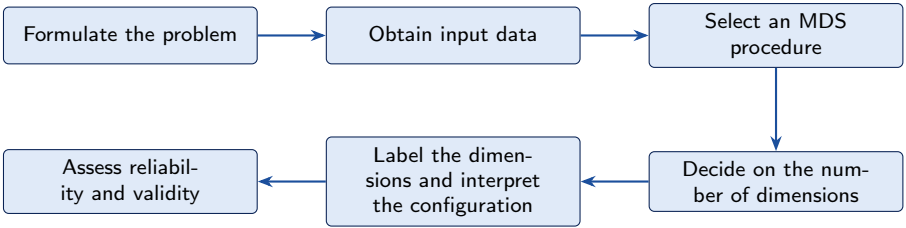
- **Image measurement**: compare how customers, non-customers and the firm itself perceive the company, and identify perceptual gaps.
- **Market segmentation**: place brands and consumers in the same space and find groups with similar perceptions.
- **New product development**: look for gaps in the map, which point to opportunities, and test new concepts against existing brands.
- **Advertising effectiveness**: check whether advertising achieved the intended positioning.
- **Pricing analysis**: compare maps built with and without price information to see what price does to perceptions.
- **Channel decisions**: judgements of how well brands fit different retail outlets can guide distribution.
- **Attitude-scale construction**: find the right dimensionality and configuration of the attitude space.

Statistics and terms associated with MDS

- **Similarity judgements**: ratings, on a Likert-type scale, of how similar each possible pair of brands is.
- **Preference rankings**: rank orderings of brands from most to least preferred.
- **Stress**: a *lack-of-fit* measure. Higher stress means a poorer fit.
- **R-square**: a squared correlation index giving the proportion of variance in the optimally scaled data that the MDS procedure accounts for. It measures *goodness of fit*.
- **Spatial map**: perceived relationships among brands shown as geometric relationships among points in a multidimensional space.
- **Coordinates**: the position of a brand or stimulus in the spatial map.
- **Unfolding**: representing *both* brands and participants as points in the same space.

Conducting MDS

Figure 2. Steps in conducting MDS



A wide variety of data can feed MDS, so the problem and the form of the data must be chosen carefully. Choosing the **number of dimensions** is the central decision in every MDS analysis.

Step 1: Formulate the problem

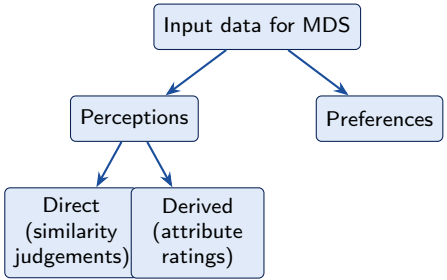
- Specify the **purpose** the MDS results will serve.
- Select the **brands or stimuli** to include:
  - at least **8** stimuli to obtain a well-defined map;
  - more than **25** is cumbersome and causes participant fatigue.
- The stimulus set shapes the dimensions that emerge. If luxury fashion brands are left out of a study of fashion brands, a "luxury" dimension may never appear.
- Base the choice on the research problem, theory and the researcher's judgement.

Running example: bottled beer

10 brands: Becks, Budvar, Budweiser, Carlsberg, Corona, Grolsch, Harp, Holsten, San Miguel and Stella Artois.

Step 2: Obtain input data

Figure 3. Input data for MDS



- **Direct approaches**: participants judge how similar or dissimilar brands are, using *their own criteria*.
  - Rate every pair on a Likert scale: $n(n-1)/2$ pairs. For 10 beers that is 45 pairs.
  - Or rank-order all pairs from most to least similar.
  - Or rank brands by similarity to an anchor brand, using each brand as the anchor in turn.
- **Derived approaches**: participants rate brands on researcher-identified attributes, and a similarity measure such as Euclidean distance is computed for each pair.

Direct approach: collecting similarity judgements

Example: similarity judgements for bottled beer brands

	Very dissimilar				*Very similar*		
Becks versus Budweiser	1	2	3	4	5	6	7
Budweiser versus Carlsberg	1	2	3	4	5	6	7
Carlsberg versus Corona	1	2	3	4	5	6	7
⋮							
Becks versus Stella Artois	1	2	3	4	5	6	7

In the beer study the direct approach was used: participants rated all 45 pairs on a seven-point scale. The next slide shows the ratings from **one participant**.

Similarity ratings of bottled beer brands

Table 26.1 Similarity ratings from one participant (7 = very similar)

	Becks	Budvar	Budweis.	Carlsb.	Corona	Grolsch	Harp	Holsten	S. Miguel	Stella
Becks										
Budvar	5									
Budweiser	6	7								
Carlsberg	4	6	6							
Corona	2	3	4	5						
Grolsch	3	3	4	4	5					
Harp	2	2	2	3	5	5				
Holsten	2	2	2	2	6	5	6			
San Miguel	2	2	2	2	6	6	7	6		
Stella Artois	1	2	4	2	4	3	3	4	3	

Only the lower triangle is needed, because MDS assumes the similarity of A to B equals the similarity of B to A.

Derived approach and how the two compare

Example: attribute ratings (semantic differential)

Best drunk with food	__:__:__:__:__:__:__	Best drunk on its own
Bottle feels good to hold	__:__:__:__:__:__:__	Bottle does not feel good to hold
Has a strong smell of hops	__:__:__:__:__:__:__	No smell of hops

Direct (similarity judgements)

- \+ No need to identify the salient attributes; participants use their own criteria.
- – The criteria depend on the brands evaluated. If all brands cost the same, price never emerges.
- – Hard to decide how to combine participants, and hard to label dimensions.

Derived (attribute ratings)

- \+ Easy to cluster participants with homogeneous perceptions.
- \+ Easier to label the dimensions.
- – The researcher must identify *all* salient attributes, and the map depends on that choice.

Direct approaches are used more often. A good practice is to use **both**: similarity judgements to build the map, attribute ratings to help interpret it.

Preference data

Preference data order brands by participants' preference for some property. They can be collected as:

- **preference rankings**, from most to least preferred;
- **paired comparisons**, indicating which brand in each pair is preferred;
- **preference ratings** for each brand.

In a map built from preference data, **distance means a difference in preference**. A preference map can look very different from a similarity map:

Example

A group of participants may see Becks and Harp as very different brands, so they lie far apart on a perception map. Yet if the two are about equally preferred, they lie close together on a preference map.

Step 3: Select an MDS procedure

The choice depends on three things:

1. Whether **perception** data, **preference** data or both are scaled.
2. The **nature of the input data**:
   - **Non-metric MDS** assumes ordinal input but produces metric output. It finds the map whose rank order of distances best reproduces the input rank order.
   - **Metric MDS** assumes metric (interval or ratio) input. Because the output is also metric, the relationship between input and output is stronger.
   - In practice the two produce similar results.
3. **Individual or aggregate** analysis:
   - Individual-level analysis gives one map per participant. It is useful for research but rarely for managers.
   - Aggregate analysis usually assumes everyone uses the same dimensions but weights them differently.

Beer example: the Table 26.1 ratings were treated as rank-ordered and scaled with a non-metric procedure, at the individual level.

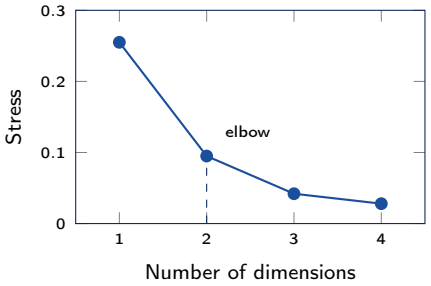
Step 4: Decide on the number of dimensions

Fit always improves as dimensions are added, so the goal is the **best fit in the fewest dimensions**. Guidelines:

1. **A priori knowledge**: theory or past research.
2. **Interpretability**: maps with more than three dimensions are hard to read.
3. **Elbow criterion**: plot stress against dimensionality and keep the number of dimensions where the curve bends sharply.
4. **Ease of use**: two-dimensional maps are easiest to work with.
5. **Statistical approaches**, for sophisticated users.

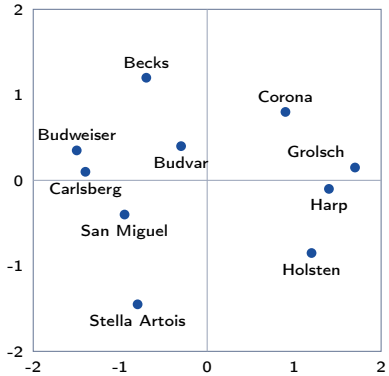
For the beer data, the elbow, interpretability and ease of use all pointed to a **two-dimensional** solution.

Figure 4. Plot of stress versus dimensionality



Two-dimensional spatial map of beer brands

Figure 5. A spatial map of beer brands



- Each point is a brand, and the **distance** between two points reflects how dissimilar the participant finds them.
- Brands close together, such as Budweiser and Carlsberg, are seen as similar and compete more directly.
- An isolated brand, such as Stella Artois, has a distinctive image.
- The axes do not have names yet. Labelling them is the next step.

Step 5: Label the dimensions and interpret the map

Labelling requires subjective judgement. Several aids help:

1. Collect ratings on researcher-supplied attributes and **fit attribute vectors** into the map, for example by regression. Label each axis after the attributes it aligns with most closely.
2. Ask participants which **criteria** they used when judging similarity.
3. Show participants their own map and ask them to **name the dimensions**.
4. Use **objective characteristics** (horsepower, fuel economy and so on) as an aid.

Reading the configuration

Nearby brands compete more fiercely. An isolated brand has a unique image. Brands further along a descriptor's direction are stronger on that characteristic. **Gaps** in the map can signal room for a new brand or a repositioning. In the beer study the axes were labelled *strength* of initial flavour and *after-taste*.

Step 6: Assess reliability and validity

MDS input data, and therefore MDS solutions, carry a lot of random variability. To check the solution:

1. Examine **R-square**. Values of **0.60 or better** are acceptable.
2. Examine **stress**, which measures badness of fit and varies with the procedure and the data.
3. For aggregate analysis, **split the data** and compare the solutions from each part.
4. **Eliminate stimuli** selectively and re-run on the remainder.
5. **Add a random error term** to the input and compare solutions.
6. Collect the data at two points in time to measure **test–retest** reliability.

Kruskal's stress formula 1

Stress (%)	Goodness of fit
20	Poor
10	Fair
5	Good
2.5	Excellent
0	Perfect

Beer example

Stress = 0.095, a **fair** fit. With Stella Artois deleted, Corona and Holsten moved somewhat, but the changes were modest, which indicates fair stability.

Assumptions and limitations of MDS

- **Symmetry**: MDS assumes the similarity of A to B equals the similarity of B to A. This can fail: New Zealand is seen as more similar to Australia than Australia is to New Zealand.
- The distance between two stimuli is assumed to be some function of their partial similarities on several perceptual dimensions. This assumption has not been tested much.
- Inter-point distances are assumed to be **ratio-scaled**, and the axes **interval-scaled**.
- Relating **physical changes** in a product to movements in the perceptual map is difficult at best.
- These limitations also apply to the scaling of preference data.

Scaling preference data: internal versus external analysis

Internal analysis of preferences

A map showing both brands and participant points (or vectors) is derived **solely from preference data**.

External analysis of preferences

Ideal points or vectors from preference data are fitted into a map derived from **perception** (similarity) data. Both kinds of data must be collected.

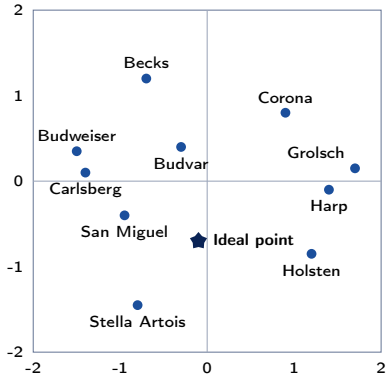
Representing brands and participants in the same space, by either method, is called **unfolding**.

External analysis is preferred in most situations, because in internal analysis:

- differences in perception are confounded with differences in preference;
- two brands seen as similar may still be preferred very differently, and internal analysis cannot show this;
- the procedures are computationally difficult.

External analysis: locating the ideal point

Figure 6. External analysis of preference data



The participant ranked the beers (most preferred first):

Stella Artois, *Holsten*, *Harp*, *San Miguel*, *Carlsberg*, *Grolsch*, *Budvar*, *Budweiser*, *Corona*, *Becks*.

These rankings and the map coordinates were entered into a preference-scaling program.

- The ideal point lies close to the most-preferred brands and far from the least preferred (Corona, Becks).
- A new brand's distance from the ideal point, relative to other brands, predicts how much it will be preferred.

Example: joint space of brands and consumers

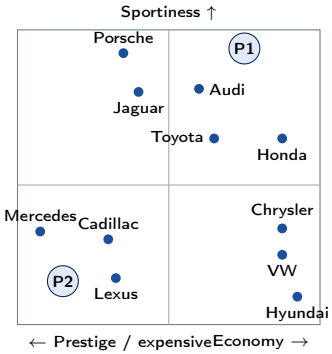
Real research: Participants park in different spaces

Car brands were mapped from similarity data on two dimensions: *economy vs prestige* and *sportiness*.

- VW and Chrysler are seen as similar. VW and Mercedes are not.
- Honda competes with Toyota, not with Porsche.
- Ideal points: **P1** prefers sporty cars, **P2** luxury cars.

Clustering ideal points **segments** the market.

Figure 7. Joint space of car brands and ideal points (illustrative)



Correspondence analysis

Correspondence analysis

An MDS technique for scaling **qualitative data**. The input is a **contingency table**. Rows and columns are scaled in corresponding units, so both can be displayed in the same low-dimensional space.

The maps show (1) similarities within the rows for a given column category, (2) similarities within the columns for a given row category, and (3) relationships between rows and columns. Interpretation resembles principal components analysis: categories that sit closer together are more similar in underlying structure.

Advantages

- Low burden on participants: they only tick which attributes apply to each brand.
- The input is simply the count of "yes" answers per brand and attribute.

Disadvantages

- Distances *between* sets (a row and a column) cannot be meaningfully interpreted.
- It is exploratory, so it is not suitable for hypothesis testing.

MDS, factor analysis and discriminant analysis

With **attribute-based** input (each participant rates n brands on m attributes), spatial maps can also be produced by two techniques covered earlier.

Factor analysis

- Factor-analyse the ratings to get a factor score for each brand on each factor.
- Plot brand scores on the factors to get a map, one per participant. Average the scores across participants for an aggregate map.
- Label dimensions using the **factor loadings**, the correlations between attributes and factors.

Discriminant analysis

- Dependent variable: the brand rated. Predictors: the attribute ratings.
- Find the linear combinations of attributes that best discriminate among brands.
- Plot the **discriminant scores** for the brands.
- Label dimensions using the **discriminant weights**.

In-class exercise: luxury car brands

One participant's ratings (1 = poor, 7 = excellent)

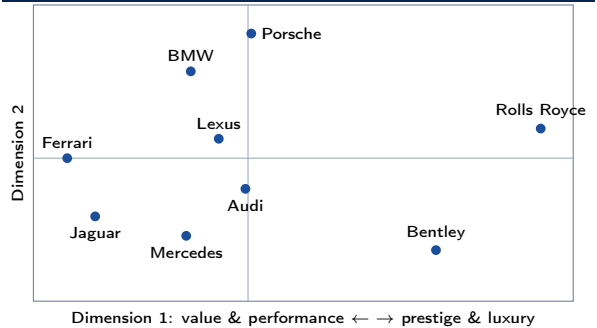
Brand	Prestige	Perform.	Luxury	Value
Ferrari	5	7	5	7
Jaguar	5	6	5	7
BMW	5	7	6	5
Mercedes	6	6	6	6
Audi	5	5	6	5
Lexus	6	6	5	5
Porsche	5	6	5	4
Bentley	7	4	7	3
Rolls Royce	7	4	7	1

Task

1. Is this a *direct* or a *derived* approach? How would you turn these ratings into similarities?
2. Produce a two-dimensional MDS map of the nine brands.
3. Label the dimensions. Which attributes line up with each axis?
4. Which brands compete most closely? Is there a gap a new brand could fill?

Exercise solution: one possible map

Figure 8. MDS map of luxury car brands



Method: Euclidean distances between the brands' rating profiles (a derived approach), scaled with classical MDS.

- **Dimension 1** explains about 86% of the variance. It correlates with luxury and prestige ($r \approx 0.83$) and negatively with value ($r \approx -0.97$) and performance ($r \approx -0.88$).
- **Dimension 2** is weak (about 8%). It separates Porsche and BMW from Mercedes, Jaguar and Bentley.
- Rolls Royce and Bentley sit apart at the prestige end. Ferrari and Jaguar cluster as performance/value brands.

Other software may rotate or reflect the map; relative positions are what matter.

Basic concepts in conjoint analysis

Conjoint analysis

A technique that determines the **relative importance** consumers attach to salient attributes and the **utilities** they attach to the levels of those attributes.

- Participants evaluate **stimuli** built from combinations of attribute levels.
- The procedure assigns values to each level so that the resulting utilities match the participants' evaluations as closely as possible.
- The underlying assumption is that any product, brand or company is evaluated as a **bundle of attributes**.

Compared with MDS: both rely on subjective evaluations. MDS stimuli are actual brands and its output is a spatial map. Conjoint stimuli are researcher-designed combinations of attribute levels, and its output is a set of **part-worth (utility) functions**.

Uses of conjoint analysis

- **Relative importance of attributes** in the choice process.
- **Market share estimation**: utilities feed a *choice simulator* that predicts the share of choices for brands with different attribute levels.
- **The most-preferred brand**: the combination of levels with the highest utility.
- **Segmentation**: cluster participants on their part-worths to find segments with similar preferences.

Applications span consumer goods, industrial goods, financial and other services: new products, competitive analysis, pricing, segmentation, advertising and distribution.

Real research: The conjoint path over the cultural divide

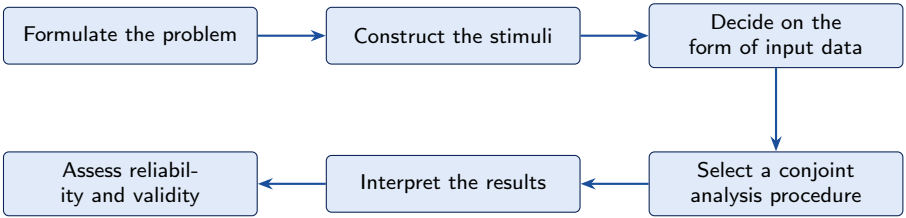
Boots was considering new stores in the Netherlands, Japan and Thailand. Conjoint analysis was used to find the key drivers of store choice (range, price, quality, service, convenience) and the trade-offs between them. Participants also rated competitors and a video of a Boots concept store on the same attributes.
With over 1,000 interviews per country, the results shaped store design, merchandising, staff training and marketing, and showed which competitors posed the biggest threat.

Statistics and terms associated with conjoint analysis

- **Part-worth functions**: also called utility functions, they describe the utility consumers attach to the levels of each attribute.
- **Relative importance weights**: show which attributes most influence consumer choice.
- **Attribute levels**: the values the attributes take.
- **Full profiles**: complete descriptions of brands on all attributes, built from the levels the design specifies.
- **Pairwise tables**: participants evaluate two attributes at a time until all required pairs are covered.
- **Cyclical designs**: reduce the number of paired comparisons.
- **Fractional factorial designs**: reduce the number of profiles in the full-profile approach.
- **Orthogonal arrays**: fractional designs that allow efficient estimation of all main effects.
- **Internal validity**: the correlation between predicted and observed evaluations of holdout (validation) stimuli.

Conducting conjoint analysis

Figure 9. Steps in conducting conjoint analysis



The salient attributes and their levels are used to build the stimuli. Participants rate or rank the stimuli, and the data are analysed, interpreted and checked for reliability and validity.

Step 1: Formulate the problem

- Attributes should be **salient** in influencing choice: for a car, price, fuel efficiency and interior space.
- They should also be **actionable**. "Sporty" only helps a manager if it is defined in attributes the manager controls.
- Sources: management and industry experts, secondary data, qualitative research and pilot surveys. A typical study uses **six or seven attributes**.
- Keep the number of levels down. More levels mean more parameters and more stimuli for participants.
- Use **at least three levels** when utility may be non-linear (for example, medium cars preferred to small or large). Binary attributes, like a sun-roof, need only two.
- The levels chosen affect evaluations, so use levels that are realistic in the market.

Table 26.2 Boot attributes and levels

Attribute	Level	Description
Uppers	1	Leather
	2	Suede
	3	Imitation leather
Country	1	Italy
	2	USA
	3	Far East
Price	1	€50
	2	€125
	3	€200

Step 2: Construct the stimuli

Full-profile approach

Also called multiple-factor evaluations. Each stimulus is a complete profile on all attributes, usually printed on its own index card.

Table 26.3 – Example boot profile

Upper: made of leather
Country: designed and made in Italy
Price: costing €200

Pairwise approach

Also called two-factor evaluations. Participants evaluate two attributes at a time, ranking every combination of their levels in a matrix (1 = most preferred, 9 = least preferred).

Example pairwise matrix: Upper × Country

	Italy	USA	Far East
Leather			
Suede			
Imitation leather			

Repeated for Upper × Price and Country × Price.

Reducing the number of stimuli

- Evaluating every combination is not necessary, and often not feasible.
  - Pairwise: **cyclical designs** reduce the number of paired comparisons.
  - Full profile: **fractional factorial designs** reduce the number of profiles. **Orthogonal arrays** estimate all main effects on an uncorrelated basis, assuming interactions are negligible.
- Two sets of data are usually collected: an **estimation set** to compute the part-worths, and a **holdout set** to assess reliability and validity.

Pairwise vs full profile

Pairwise judgements are easier to give, but they require *more* evaluations and can feel unrealistic. Both yield comparable utilities. Full profile is more common.

Boot example

3 attributes $\times$ 3 levels gives $3^3 = 27$ profiles. A fractional factorial design reduced this to **9 estimation profiles**, plus 9 validation profiles.

Step 3: Decide on the form of input data

- **Non-metric**: participants **rank** the profiles (or the cells of each pairwise matrix). Proponents argue that rankings mirror real marketplace behaviour.
- **Metric**: participants **rate** each profile independently. Advocates find ratings more convenient for participants and easier to analyse. Ratings have become increasingly common.
- The dependent variable is usually **preference** or **intention to buy**, but the method can also handle actual purchase or choice.

Boot example: ratings on a 9-point Likert scale (1 = not preferred, 9 = greatly preferred).

Table 26.4 Boot profiles and their ratings

Profile	Upper	Country	Price	Rating
1	1	1	1	9
2	1	2	2	7
3	1	3	3	5
4	2	1	2	6
5	2	2	3	5
6	2	3	1	6
7	3	1	3	5
8	3	2	1	7
9	3	3	2	6

Levels as in Table 26.2.

Step 4: The conjoint analysis model

The basic model writes the overall utility of an alternative as the sum of its part-worths:

$$U(X) = \sum_{i=1}^{m} \sum_{j=1}^{k_i} \alpha_{ij}\, x_{ij}$$

- $U(X)$: overall utility of an alternative
- α_{ij}: part-worth of level j of attribute i
- $x_{ij} = 1$ if level j of attribute i is present, 0 otherwise
- k_i: number of levels of attribute i; m: number of attributes

Attribute importance

Importance is the range of the part-worths:

$$I_i = \max_j(\alpha_{ij}) - \min_j(\alpha_{ij})$$

normalised into a relative weight:

$$W_i = \frac{I_i}{\sum_{i=1}^{m} I_i}, \qquad \sum_{i=1}^{m} W_i = 1$$

Estimation with dummy-variable regression

- The simplest and increasingly popular method is **OLS regression with dummy variables**.
- An attribute with k_i levels is coded with $k_i - 1$ dummies. Level 3 is the base level.
- Metric data: ratings are the dependent variable. Non-metric data: rankings become 0/1 via paired comparisons, or use LINMAP, MONANOVA or LOGIT.
- For aggregate analysis, cluster participants on their part-worths and analyse each cluster.

Uppers	X_1	X_2
Level 1 (leather)	1	0
Level 2 (suede)	0	1
Level 3 (imitation)	0	0

Table 26.5 Boot data coded for regression

	Upper		Country		Price	
Y	X_1	X_2	X_3	X_4	X_5	X_6
9	1	0	1	0	1	0
7	1	0	0	1	0	1
5	1	0	0	0	0	0
6	0	1	1	0	0	1
5	0	1	0	1	0	0
6	0	1	0	0	1	0
5	0	0	1	0	0	0
7	0	0	0	1	1	0
6	0	0	0	0	0	1

$$\hat{U} = b_0 + b_1 X_1 + b_2 X_2 + \cdots + b_6 X_6$$

From regression coefficients to part-worths

Estimates: $b_0 = 4.222,\ b_1 = 1.000,\ b_2 = -0.333,\ b_3 = 1.000,\ b_4 = 0.667,\ b_5 = 2.333,\ b_6 = 1.333$

Each dummy coefficient is the part-worth of that level *minus* the part-worth of the base level. Utilities are interval-scaled, so their origin is arbitrary. Fixing it requires one more constraint: **the part-worths of each attribute sum to zero**.

Uppers

$$\alpha_{11} - \alpha_{13} = 1.000$$
$$\alpha_{12} - \alpha_{13} = -0.333$$
$$\alpha_{11} + \alpha_{12} + \alpha_{13} = 0$$
$$\Rightarrow \alpha_{11} = 0.778$$
$$\alpha_{12} = -0.556$$
$$\alpha_{13} = -0.222$$

Country

$$\alpha_{21} - \alpha_{23} = 1.000$$
$$\alpha_{22} - \alpha_{23} = 0.667$$
$$\alpha_{21} + \alpha_{22} + \alpha_{23} = 0$$
$$\Rightarrow \alpha_{21} = 0.445$$
$$\alpha_{22} = 0.111$$
$$\alpha_{23} = -0.556$$

Price

$$\alpha_{31} - \alpha_{33} = 2.333$$
$$\alpha_{32} - \alpha_{33} = 1.333$$
$$\alpha_{31} + \alpha_{32} + \alpha_{33} = 0$$
$$\Rightarrow \alpha_{31} = 1.111$$
$$\alpha_{32} = 0.111$$
$$\alpha_{33} = -1.222$$

Shortcut: with three levels, the base-level part-worth is $\alpha_{i3} = -(b_{i1} + b_{i2})/3$. For uppers: $-(1.000 - 0.333)/3 = -0.222$.

Results of the conjoint analysis

Table 26.6 Results of conjoint analysis

Attribute	Level	Utility	Importance
Uppers	Leather	0.778	
	Suede	−0.556	0.286
	Imitation leather	−0.222	
Country	Italy	0.445	
	USA	0.111	0.214
	Far East	−0.556	
Price	€50	1.111	
	€125	0.111	0.500
	€200	−1.222	

Relative importance weights

Sum of part-worth ranges:
$1.334 + 1.001 + 2.333 = 4.668$

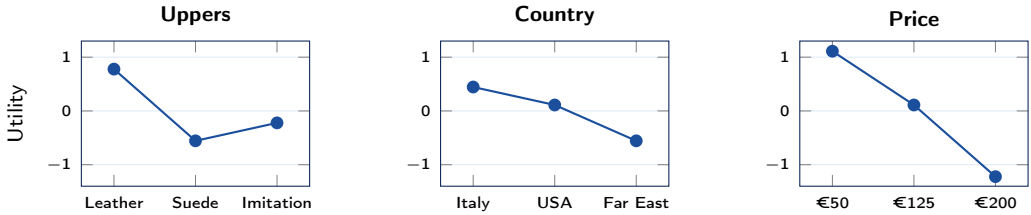
- Uppers: $1.334/4.668 = 0.286$
- Country: $1.001/4.668 = 0.214$
- Price: $2.333/4.668 = 0.500$

Check

Predicted utility of profile 1 (leather, Italy, €50): $4.222 + 1.000 + 1.000 + 2.333 = 8.56$, against an observed rating of 9. The regression R^2 is 0.934.

Step 5: Interpret the results

Figure 10. Part-worth functions for the boots example



Leather uppers are most preferred, then imitation leather, with suede last. Italian boots are preferred to American, which are preferred to those from the Far East. €50 has the highest utility and €200 the lowest. **Price is the most important attribute (0.50)**, so this participant can be labelled *price sensitive*. Utilities are interval-scaled, so only differences between levels are meaningful.

Step 6: Assess reliability and validity

1. **Goodness of fit** of the estimated model: with dummy-variable regression, look at R^2. Models with poor fit are suspect.
2. **Test–retest reliability**: later in the interview, ask participants to evaluate some stimuli again and correlate the two sets of answers.
3. **Internal validity**: use the part-worths to predict evaluations of the **holdout** stimuli and correlate the predictions with the actual evaluations.
4. **Stability**: for aggregate analysis, split the sample in several ways and compare the results across subsamples.

Boot example

$R^2 = 0.934$, a good fit. Predicted and actual ratings of the nine validation profiles correlate at $r = 0.95$ (significant at $\alpha = 0.05$), which shows good predictive ability.

Assumptions and limitations of conjoint analysis

- It assumes the **important attributes can be identified**.
- It assumes consumers evaluate alternatives **in terms of these attributes** and make **trade-offs**. Where image or brand name dominates, consumers may not think in attributes at all.
- Even when they do, the **trade-off model** may not represent the actual choice process well.
- **Data collection can be complex**, especially with many attributes and individual-level models. Adaptive and hybrid conjoint procedures ease this.
- The part-worth functions are **not unique**: they are interval-scaled, with an arbitrary origin.

Hybrid conjoint analysis

Hybrid conjoint analysis

A form of conjoint analysis that **simplifies the data-collection task** and can estimate **selected interactions** as well as all main effects.

In traditional conjoint, each participant evaluates many profiles, yet usually only main effects are estimated, although attributes may *interact*. Hybrid models address both problems:

- Each participant evaluates a limited number of full profiles, **generally no more than nine**.
- The profiles come from a large **master design**. Different participants see different sets, so the group as a whole covers every profile of interest.
- Participants also **directly rate** the importance of each attribute and the desirability of each level.
- Combining the direct and the conjoint evaluations allows an aggregate model that still keeps some individual differences.

Using MDS and conjoint together

Real research: Weeding out the competition

ICI Agricultural Products (now part of AkzoNobel) was unsure whether to cut the price of **Fusilade**, its herbicide. It interviewed 601 soybean and cotton farmers face to face.

- **Conjoint analysis** measured the relative importance of the attributes farmers use to choose herbicides.
- **MDS** mapped farmers' perceptions of the competing herbicides.

Price mattered a great deal, especially above €10 per hectare. But farmers would also pay more for good weed control: a herbicide that failed to control even one of the four most common weeds had to be very cheap to win share. Fusilade promised good control, and MDS showed that a key competitor was seen as expensive.

Decision: ICI kept its original pricing.

More applications

Real research: Fab's fabulous foamy fight

Colgate entered Thailand's superconcentrate detergent market with Fab Power Plus, targeting a 4% share. Secondary data and qualitative research identified the critical factors: environmental appeal, hand- and machine-washing convenience, cleaning ability, the right level of suds and brand name.

A conjoint study varied these factors at two or three levels. Part-worths, estimated at the individual and group level, showed that **hand–machine capability** contributed substantially. No existing brand offered both, so Fab was launched with both capabilities.

Real research: Ethical perceptions of research firms

MDS can plot the perceived ethics of marketing research firms on two dimensions: *broad-based moral equity* and *relativism*.

In the illustrative map, internal research departments were seen as the most ethical on both dimensions. Large firms scored higher on relativism and small firms on moral equity. Full-service firms scored above limited-service firms on both.

Software: running the analyses in SPSS

MDS (ALSCAL): `Analyze > Scale > Multidimensional Scaling`

1. Convert similarities to distances by subtracting each value of Table 26.1 from 8. The matrix must be square symmetric, with zeros on the diagonal.
2. Move the brand variables into the VARIABLES box.
3. Under DISTANCES, check *Data are distances*, shape *Square symmetric*.
4. MODEL: level of measurement *Interval*, scaling model *Euclidean distance*, conditionality *Matrix*.
5. OPTIONS: display *Group plots*, *Data matrix* and *Model and options summary*.

ALSCAL handles ordinal, interval or ratio data, direct or derived approaches, and individual-differences or aggregate analysis.

Conjoint analysis

- With a metric dependent variable, use dummy-variable regression: `Analyze > Regression > Linear`.
- Dependent variable: the rating. Predictors: the $k_i - 1$ dummies for each attribute.
- **SPSS Conjoint** is a specialised module, sold separately.

Note

Different programs may give MDS solutions that are rotated or reflected relative to the textbook's. Interpret relative positions, not signs.

Summary

Multidimensional scaling

- Represents perceptions and preferences as points in a spatial map.
- Perception data are direct (similarities) or derived (attribute ratings). Direct approaches are more common.
- The procedure depends on metric vs non-metric input, perceptions vs preferences, and individual vs aggregate analysis.
- Choose dimensions using theory, interpretability, the elbow criterion and ease of use.
- Labelling dimensions requires judgement. Check R^2 (0.60 or better), stress and stability.
- Preference data: internal or external analysis. External is preferred.
- Qualitative data: correspondence analysis. Factor and discriminant analysis can also produce maps.

Conjoint analysis

- Estimates attribute importance and level utilities from evaluations of profiles.
- Choose salient, actionable attributes and realistic levels.
- Stimuli are built with the pairwise or full-profile approach. Fractional designs reduce the workload.
- Input is rankings (non-metric) or ratings (metric). The dependent variable is usually preference or intention to buy.
- Dummy-variable regression is increasingly the method of choice.
- Interpret the part-worth functions and importance weights, then assess fit, test–retest reliability and holdout validity.
- Hybrid conjoint reduces the burden on each participant.

Review questions

1. For what purposes are MDS procedures used?
2. Identify two marketing research problems where MDS could be applied, and explain how you would apply it.
3. What is meant by a spatial map?
4. Describe the steps involved in conducting MDS.
5. Describe the direct and derived approaches to obtaining MDS input data.
6. What factors influence the choice of an MDS procedure?
7. What guidelines are used to decide the number of dimensions?
8. How can the reliability and validity of MDS solutions be assessed?
9. What is the difference between internal and external analysis of preference data?
10. What is involved in formulating a conjoint analysis problem?
11. Describe the full-profile approach to constructing stimuli.
12. Describe the pairwise approach to constructing stimuli.
13. How can regression analysis be used to analyse conjoint data?
14. Illustrate graphically what is meant by part-worth functions.
15. What procedures are available to assess the reliability and validity of conjoint results?
16. Identify two problems where conjoint analysis could be applied, and explain how you would use it.